Dissertation Defense: Essays in Bayesian Inference and Deep Learning

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My dissertation will include papers in both Bayesian inference and deep learning area.

- Deep Learning in Characteristics-Sorted Factor Models
- Weighted Bayesian Bootstrap for Scalable Posterior Distributions
- Bayesian Inference for Pólya Inverse Gamma Models

In this presentation, I will only talk about the last paper.

Motivation

Data Augmentation

Pólya Inverse Gamma (P-IG) Distribution

Applications

- Gamma Shape
- Concentration Parameter of Multinomial-Dirichlet Model

Discussion

Bayesian Inference for Gamma Models

- We propose a framework of data augmentation strategy which can be applied to a variety of models:
  - Negative binomial regression for count data
  - Topic models in natural language processing
  - Generalized Pareto distribution in extreme value analysis
  - ...
- Coauthors: Nicholas Polson and Jingyu He.

Gamma-Related Distributions

Gamma-related distributions are key elements of many statistical models.

- Gamma distribution (exponential and Poisson rate, λ):

$$p(x|\alpha,\beta) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}$$

- Inverse gamma distribution (normal variance, σ^2):

$$p(x|\alpha,\beta) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{-\alpha-1} e^{-\beta/x}$$

- Beta distribution (binomial and geometric probability, p):

$$p(x|\alpha,\beta) = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1}(1-x)^{\beta-1}$$

- Dirichlet distribution (multinomial probability vector, p):

$$p(x|\boldsymbol{\alpha}) = \frac{\Gamma(\sum_i \alpha_i)}{\prod_i \Gamma(\alpha_i)} \prod_{i=1}^{K} x_i^{\alpha_i-1}$$

Technical Difficulties

- The normalizing constants of these distributions depend on gamma functions whose arguments include shape (gamma, inverse gamma) and concentration (beta, Dirichlet) parameters.
- Bayesian learning of parameters nested inside the gamma function presents significant technical difficulties since there is no known conjugate prior distribution.
- Gamma shape example:

$$\text{Likelihood function: } p(x|\alpha,\beta) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}$$

How to sample from the posterior of α, $p(x|\alpha)p(\alpha)$?

$$p(\alpha|\beta,x) \propto \frac{(\beta x)^\alpha}{\Gamma(\alpha)} p(\alpha)$$

Existing Approaches

- Metropolis–Hastings, rejection sampling, etc.
  - Acceptance rate is low. Converge slowly.
- Approximate the posterior with gamma distribution:
  - Stirling's formula. Match mode. (Rossell et al., 2009)
  - Gamma prior. Match derivatives. (Miller, 2018)
- Why are we not satisfied with the approximation approach?
  - Derived only for 1-d gamma shape parameter. For other models, gamma approximation doesn't generalize well.

$$\frac{(\beta x)^{\alpha}}{\Gamma(\alpha)} \approx \alpha^{c-1} e^{-(\tau - \log(\beta x))\alpha},$$ But $\Gamma(\alpha)$ grows much faster than $\alpha^{1-c} e^{\tau\alpha}$!

  - Need extra modelling for correlations in multidimensional case.
  - Newton–Raphson steps to solve for the approximating gamma parameters. Can still be burdensome for high-dimensional cases, where there is a large system of nonlinear equations.

Data Augmentation Strategy

- Goal: sample the random vector $\boldsymbol{\alpha} \in \mathbb{R}^d$ from a posterior $p(\boldsymbol{\alpha}|x)$, which usually has the form

$$p(\boldsymbol{\alpha}|x) \propto \frac{1}{\Gamma(b_1'\boldsymbol{\alpha} + c_1)} \frac{1}{\Gamma(b_2'\boldsymbol{\alpha} + c_2)} \cdots \frac{1}{\Gamma(b_K'\boldsymbol{\alpha} + c_K)} h(\boldsymbol{\alpha}, x)$$

where $b_1, b_2, \ldots b_K \in \mathbb{R}^d$ and $c_1, c_2, \ldots c_K \in \mathbb{R}$ are functions of x.

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- Solution:
  - introduce auxiliary variables ω's with a special distribution.
  - sample the augmented random vector $(\boldsymbol{\alpha}, \boldsymbol{\omega})$ from the joint distribution $p(\boldsymbol{\alpha}, \boldsymbol{\omega}|x)$.
  - discard $\boldsymbol{\omega}$ samples and keep $\boldsymbol{\alpha}$ samples.

$$p(\boldsymbol{\alpha}|x) = \int p(\boldsymbol{\alpha}, \boldsymbol{\omega}|x) d\omega$$

How Does It Work?

- If we can write out the following integral identity, with $p(\alpha|\omega, x)$ and $p(\omega|x)$ being some well-known probability densities,

$$p(\alpha|x) = f(\alpha, x)/Z_x = \int p(\alpha|\omega, x)p(\omega|x)d\omega,$$

then it's a two-step sampling: first sample $\omega|x$, then sample $\alpha|\omega, x$. However, the normalizing constant Z_x is hard to compute.

- Work around: find two nonnegative functions $g(\alpha, \omega, x)$ and $q(\omega)$

$$f(\alpha, x) = \int g(\alpha, \omega, x)q(\omega)d\omega,$$

$$p(\alpha|x) = \int g(\alpha, \omega, x)q(\omega)/Z_x d\omega$$

$$Z_x = \int g(\alpha, \omega, x)q(\omega)d\alpha d\omega$$

Hence $g(\alpha, \omega, x)q(\omega)/Z_x$ is the joint density of (α, ω), conditioned on x.

Gibbs Sampling

- In order to sample from the joint $p(\alpha, \omega|x) := g(\alpha, \omega, x)q(\omega)/Z_x$, we don't actually need to know the value of Z_x.
- Gibbs sampler:
  1. Sample $\alpha|\omega, x$ from
  $$p(\alpha|\omega, x) = \frac{g(\alpha, \omega, x)q(\omega)}{\int g(\alpha, \omega, x)q(\omega)d\alpha} \propto g(\alpha, \omega, x)$$
  2. Sample $\omega|\alpha, x$ from
  $$p(\omega|\alpha, x) = \frac{g(\alpha, \omega, x)q(\omega)}{\int g(\alpha, \omega, x)q(\omega)d\omega} \propto g(\alpha, \omega, x)q(\omega)$$
  3. Require: $g(\alpha, \omega, x)$ as a (unnormalized) density of α is easy to sample; the same for $g(\alpha, \omega, x)q(\omega)$ as a density of ω.
     - We show that $g(\alpha, \omega, x) = h(\alpha)e^{\eta(\alpha)'T(\omega,x)}$, i.e. exponential family.
     - $g(\alpha, \omega, x)q(\omega)$ and $q(\omega)$ are "conjugate", i.e. same distribution class.

Back to Gamma Function

Useful integral identities:

- $$\Gamma(\alpha) = \int_0^\infty \omega^{\alpha-1} e^{-\omega} d\omega$$

  Here $g(\alpha, \omega) = e^{\log(\omega)*\alpha}$, $q(\omega) = \omega^{-1} e^{-\omega}$. Both $q(\omega)$ and $g(\alpha, \omega) q(\omega)$ are gamma densities for ω, with shape parameter 0 and α.

- What about the reciprocal gamma function?

$$\frac{1}{\Gamma(\alpha)} = \int ??? d\omega$$

Weierstrass's Definition of Gamma Function

The definition for the gamma function due to Weierstrass is valid for all complex numbers α except the non-positive integers:

$$\frac{1}{\Gamma(\alpha)} = \alpha e^{\gamma\alpha} \prod_{k=1}^{\infty} \left(1 + \frac{\alpha}{k}\right) e^{-\frac{\alpha}{k}}$$

where $\gamma = -\psi(1) \approx 0.577$ is the Euler-Mascheroni constant and $\psi(x) = \frac{d}{dx} \log \Gamma(x)$ is called digamma function.

This inspires us to define a new class of distributions.

Pólya Inverse Gamma (P-IG) Distribution

- Definition (Pólya Inverse Gamma)

Random variable ω has a Pólya inverse Gamma distribution, P-IG(0), if and only if its density $p(\omega) = q(\omega)/Z$ satisfies the following identity

$$\int_0^\infty e^{-\omega t^2} q(\omega) d\omega = \prod_{k=1}^{\infty} \left(1 + \frac{t}{k}\right) e^{-\frac{t}{k}}.$$

for any $t \geq 0$.

- Definition (Exponential Tilting)

Furthermore, the density of P-IG(c) distribution is defined as an exponential tilting of the P-IG(0) density,

$$p_c(\omega) \propto \exp\left(-c^2\omega\right) p(\omega), \, \omega > 0.$$

Reciprocal Gamma as Mixture of Normals

- A scale mixture of normal density:

$$\frac{e^{-\gamma\alpha}}{\Gamma(\alpha+1)} = \int_0^\infty e^{-\omega\alpha^2} q(\omega) d\omega$$

- A mean-variance mixture of normal density:

$$\frac{\beta^\alpha}{\Gamma(\alpha+1)} = \int_0^\infty e^{-\omega\alpha^2+(\gamma+\log\beta)\alpha} q(\omega) d\omega$$

- A power tilted mean-variance mixture of normal density:

$$\frac{\beta^\alpha}{\Gamma(\alpha)} = \int_0^\infty \alpha e^{-\omega\alpha^2+(\gamma+\log\beta)\alpha} q(\omega) d\omega$$

- $$\begin{aligned}\frac{1}{\Gamma(\alpha)} &= \alpha e^{\gamma\alpha} \prod_{k=1}^{\infty} \left(1 + \frac{\alpha}{k}\right) e^{-\frac{\alpha}{k}},\ \alpha \in \mathbb{C}/\{0, -1, -2, ...\} \\ &= \int_0^{\infty} \alpha e^{\gamma\alpha} e^{-\omega\alpha^2} q(\omega) d\omega \\ &:= \int_0^{\infty} g(\alpha, \omega) q(\omega) d\omega\end{aligned}$$
- $g(\alpha, \omega) = \alpha e^{-\omega\alpha^2 + \gamma\alpha}$. (We will see a more general version later.)
- Both $q(\omega)$ and $g(\alpha, \omega)q(\omega)$ are (unnormalized) P-IG densities for ω, with parameter 0 and α.

Relation to Generalized Inverse Gaussian

- Theorem

The P-IG(c) class of distributions can be constructed as an infinite sum of generalized inverse Gaussian (GIG) distributions[1] *as follows*

$$P\text{-}IG(c) \stackrel{d}{=} \sum_{k=1}^{\infty} GIG\left(-\frac{3}{2}, 2c^2, \frac{1}{2k^2}\right).$$

- If $\omega_k \sim GIG\left(-\frac{3}{2}, 2c^2, \frac{1}{2k^2}\right)$,
  - Its density function $p(x) \propto x^{-\frac{5}{2}} \exp\left\{-c^2 x - \frac{1}{2k^2 x}\right\}, x > 0$
  - $\int_0^\infty e^{-t^2 x} p(x) dx = \left(1 + \frac{t}{k}\right) e^{-\frac{t}{k}}$
  - If $c = 0$, it reduces to inverse gamma distribution.
  - $E[\omega_k] = \frac{1/2}{k^2 + ck}$, $Var[\omega_k] = \frac{1/(4c)}{k^3 + 2ck^2 + c^2 k}$

[1] Barndorff-Nielsen and Halgreen (1977)

Mean and Variance

- If $\omega \sim$ P-IG(c), then

$$E(\omega) = \frac{\psi^{(0)}(c+1) - \psi^{(0)}(1)}{2c},$$
$$Var(\omega) = \frac{\psi^{(0)}(c+1) - \psi^{(0)}(1) - c\psi^{(1)}(c+1)}{4c^3}$$

where $\psi^{(n)}(\cdot)$ is the n-th derivative of digamma function.

This helps us to develop the EM algorithm.

Generate P-IG Random Variables

To approximate a P-IG random variable, the exact form of its density is not necessary.

- Approximate a P-IG random variable by the finite sum of GIG random variables. (If needed, can add a normal variable to approximate the tail part and reduce bias.)
- Generate GIG: Hörmann and Leydold (2013). R package: `GIGrvg`
- Approximate the sum of n i.i.d. P-IG:

$$\sqrt{n}\left(\left(\frac{1}{n}\sum_{i=1}^{n}\omega_i\right) - E[\omega]\right) \overset{d}{\Rightarrow} N(0, Var[\omega])$$

Applications

- Gamma Shape
- Negative Binomial Regression
- Multinomial-Dirichlet Model

Gamma Shape

A gamma distribution with shape parameter $\alpha > 0$ and rate parameter $\beta > 0$ has probability density function

$$f(y|\alpha, \beta) = \frac{\beta^\alpha}{\Gamma(\alpha)} y^{\alpha-1} e^{-\beta y}, y > 0$$

Damsleth(1975) discussed conjugate priors for the shape parameter in Gamma distribution, i.e. gamcon distributions of type I and type II.

- Type I assumes the rate parameter is known. (see example in Section 4 of our arXiv paper)
- Type II assumes the rate parameter is unknown.

$$p(x|\mu, \delta) = C \cdot \frac{\Gamma(\delta x + 1)}{\Gamma(x)^\delta} (\delta\mu)^{-\delta x}, \ \mu > 1, \delta > 0$$

We consider the hierarchical model

$$\begin{aligned} Y|\alpha,\beta &\sim \Gamma(\alpha,\beta), \\ \beta|\alpha &\sim \Gamma(\delta\alpha+1,\delta\eta), \\ \alpha &\sim \xi_2(\eta/\mu,\delta). \end{aligned}$$

Here ξ_2 is gamcon distribution of type II. $\eta > \mu > 0, \delta > 0$.

Posteriors

Let $y_1, y_2, ..., y_n$ be independently and identically distributed Gamma random variables drawn from the model. Then the marginal posterior distribution of α is also ξ_2,

$$\alpha | y_1, y_2, ..., y_n \sim \xi_2(\eta'/\mu', \delta')$$

with

$$\begin{aligned}
\delta' &= \delta + n \\
\eta' &= \frac{\delta}{\delta + n}\eta + \frac{n}{\delta + n} y_a, \\
\mu' &= \mu^{\frac{\delta}{\delta+n}} \cdot y_g^{\frac{n}{\delta+n}}.
\end{aligned}$$

y_a is the arithmetical mean of y, $y_a = \frac{1}{n}\sum_{i=1}^{n} y_i$;
y_g is the geometrical mean of y, $y_g = \left(\prod_{i=1}^{n} y_i\right)^{1/n}$.

Sampling from ξ_2 when $\delta \in \mathbb{N}^+$

Bayesian inference requires sampling from gamcon distribution.

$$p(x|\mu,\delta) = C \cdot \frac{\Gamma(\delta x+1)}{\Gamma(x)^\delta}(\delta\mu)^{-\delta x}$$
$$= C \int x^\delta \exp\left(-\tau - \left(\sum_{i=1}^{\delta}\omega_i\right)x^2 + \delta(\gamma - \log(\delta\mu/\tau))x\right)\prod_{i=1}^{\delta} p^{PIG}(\omega_i)d\tau d\boldsymbol{\omega}$$

where p^{PIG} is P-IG(0) density. δ can be interpreted as sample size.

Conditional posteriors:

- $\tau|x,\boldsymbol{\omega} \sim \Gamma(\delta x+1,1)$
- $\omega_1,\omega_2,...,\omega_\delta|x,\tau \overset{i.i.d.}{\sim}$ P-IG(x)
- $x|\tau,\boldsymbol{\omega} \sim H\left(\delta+1, \sum_{i=1}^{\delta}\omega_i, \delta\left(\gamma - \log(\delta\mu/\tau)\right)\right)$[2]

[2] H(m,a,b) is the distribution whose probability density is proportional to $x^{m-1}e^{-ax^2+bx}$ with $a>0, b\neq 0$. It can be easily sampled by rejection sampling.

Damsleth's Example

- $n = 30, y_a = 5.09, y_g = 4.26.$
- $\delta = 0$ so that the "prior sample size" is 0.
- The posterior of α is

$$\alpha | y_1, y_2, ..., y_n \sim \xi_2(y_a/y_g, n)$$

	Mean	Variance	Skewness	Kurtosis
Numerical Integration	3.252	0.585	0.490	3.361
Gibbs Sampling	3.288	0.612	0.530	3.358

Table 1: Posterior Moments

EM Algorithm

Use EM algorithm to find the posterior mode:

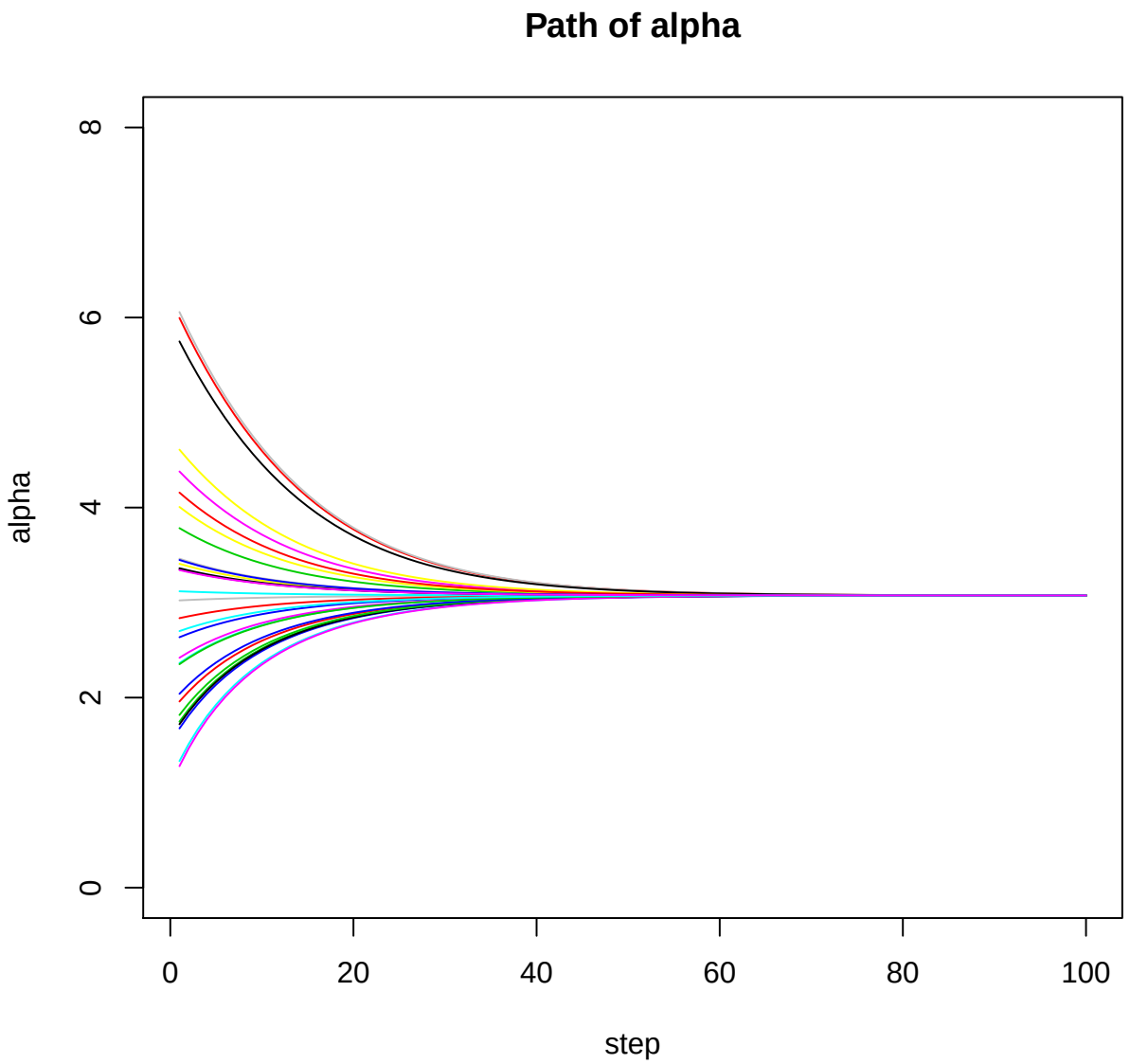
- Expectation step:

$$
\begin{aligned}
Q(\alpha|\alpha^{(t)}) &:= E_{\boldsymbol{\theta}|\boldsymbol{Y},\alpha^{(t)}}\left[\log L(\alpha;\boldsymbol{\theta},\boldsymbol{Y})\right] \\
&= c + \kappa_1 \log\alpha - \kappa_2\alpha^2 + \kappa_3\alpha \\
\kappa_1 &:= \delta' > 0 \\
\kappa_2 &:= \frac{\delta'\left(\gamma + \psi(\alpha^{(t)}+1)\right)}{2\alpha^{(t)}} > 0 \\
\kappa_3 &:= \delta'\left(\gamma - \log\delta'\eta'/\mu' + \psi(\delta'\alpha^{(t)}+1)\right)
\end{aligned}
$$

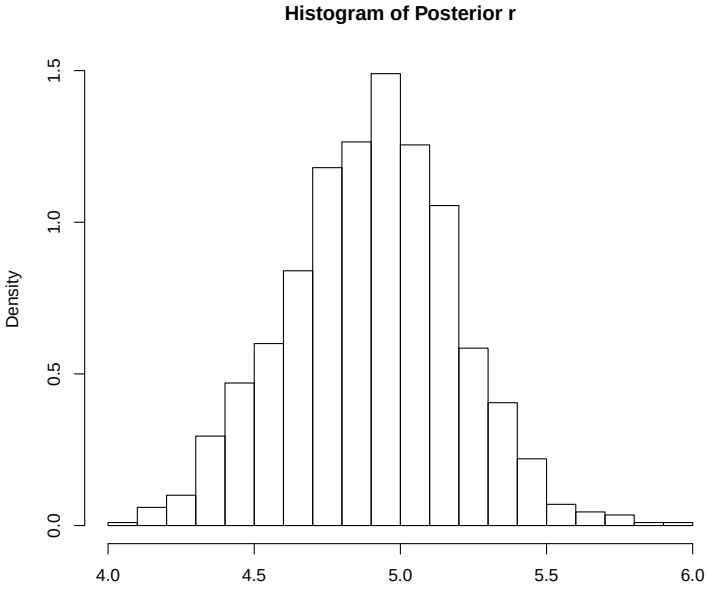
- Maximization step:

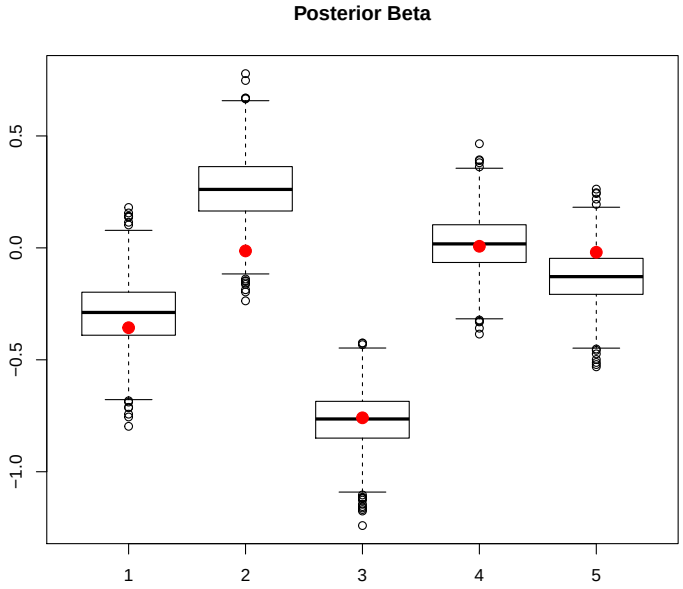
$$
\begin{aligned}
\alpha^{(t+1)} &:= \arg\max_{\alpha} Q(\alpha|\alpha^{(t)}) \\
&= \frac{\kappa_3 + \sqrt{\kappa_3^2 + 8\kappa_1\kappa_2}}{4\kappa_2} \qquad \text{(unique solution)}
\end{aligned}
$$

Simulation: $\alpha = 3, \beta = 1.5, n = 500$. Prior $\delta = 0$. Start with 30 different initial values.



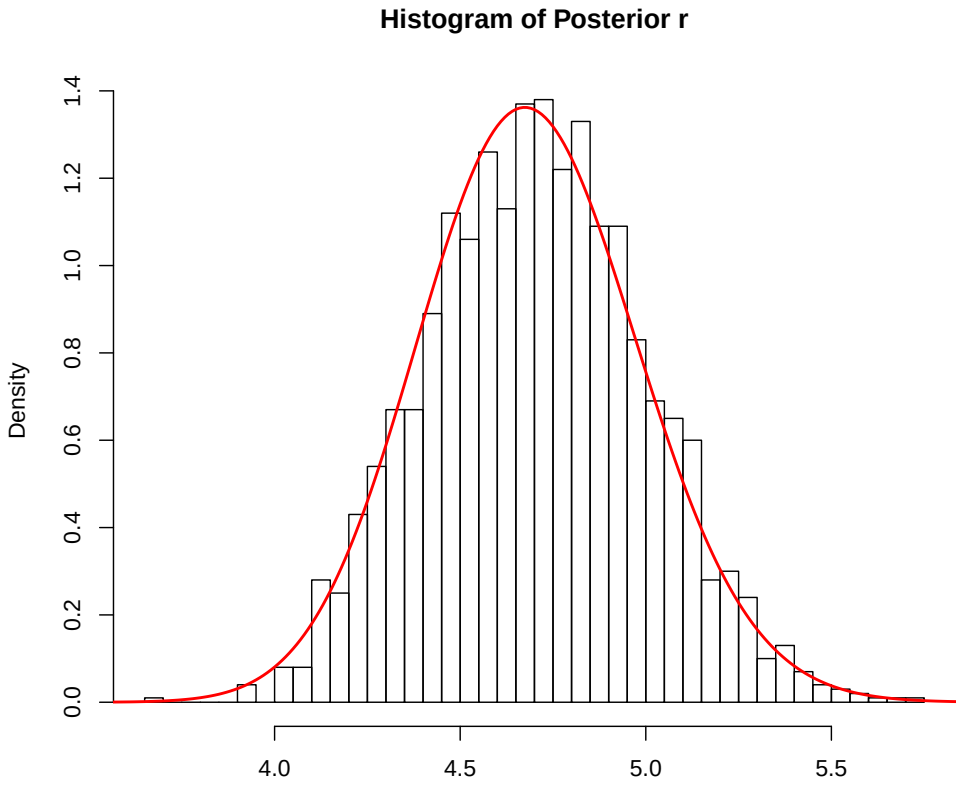
Simulation: $r = 5, n = 100, k = 5$. $x_{ij} \overset{i.i.d.}{\sim} N(0, 0.5^2)$.
Prior: $p(r) \sim 1/r, p(\beta) \sim N(0, 10^6 I_k)$.





The red dots are true β's.

Using the same dataset, but now with β known (so that it's possible to numerically calculate the posterior of r).



The red line is the target density (constant term calculated by numerical integration)

Multinomial-Dirichlet Model

- For a random customer from region m, we assume the probability that he/she chooses the phone k is p_{km}.
- For each region, we have the probability vector $\boldsymbol{p}_m = (p_{m1,\ldots,p_{mK}})$ and the observation vector $\boldsymbol{n}_m = (n_{m1}, \ldots, n_{mK})$. $N_m := \sum_{k=1}^{K} n_{mk}$.
- The distribution of $\boldsymbol{n}_m$ given N_m and $\boldsymbol{p}_m$ is multinomial:

$$\boldsymbol{n}_m | N_m, \boldsymbol{p}_m \sim \text{Multinomial}(N_m, \boldsymbol{p}_m)$$

- The probability vector $\boldsymbol{p}_m$ is assigned a Dirichlet distribution with concentration parameter vector $\boldsymbol{\alpha} = (\alpha_1, \ldots \alpha_K)$ where $\alpha_k > 0$.

$$\boldsymbol{p}_m | \boldsymbol{\alpha} \sim \text{Dirichlet}(\boldsymbol{\alpha})$$

Posteriors

Under the independent gamma prior $p(\alpha_k) \propto \alpha_k^{\tau-1} e^{-\beta\alpha_k}$,

$$\eta_1, \eta_2, .., \eta_M \mid \boldsymbol{\alpha}, \boldsymbol{p}, \boldsymbol{\omega}, \boldsymbol{n} \overset{i.i.d.}{\sim} \Gamma(\alpha_0, 1),$$
$$\omega_{1k}, \omega_{2k}, ..., \omega_{Mk} \mid \boldsymbol{\alpha}, \boldsymbol{p}, \boldsymbol{\eta}, \boldsymbol{n} \overset{i.i.d.}{\sim} \text{P-IG}(\alpha_k), k = 1, 2, ..., K$$
$$\boldsymbol{p}_m \mid \boldsymbol{\alpha}, \boldsymbol{\omega}, \boldsymbol{\eta}, \boldsymbol{n} \sim \text{Dirichlet}\left(n_{m1} + \alpha_1, \cdots, n_{mK} + \alpha_K\right)$$
$$\alpha_k \mid \boldsymbol{p}, \boldsymbol{\omega}, \boldsymbol{\eta}, \boldsymbol{n} \sim H(M + \tau, a_k, b_k), k = 1, 2, ..., K$$

where $\alpha_0 = \sum_{k=1}^{K} \alpha_k$.

$$a_k = \sum_{m=1}^{M} \omega_{mk}$$

$$b_k = M\gamma - \beta + \sum_{m=1}^{M} \log(\eta_m p_{mk})$$

Simulation: $N_m = 500$ for each m. Prior: $\alpha_k \sim 1/\alpha_k$.

- Setting A (heterogeneous): $\alpha_k = k/K$.
  Setting B (homogeneous): $\alpha_k = 1/K$.
- 500 Posterior Samples. Data augmentation with P-IG v.s. Metropolis-Hasting.
- Speed up Gibbs sampler[3]: Replace (a_k, b_k) with their expectations so that there is no need to sample those auxiliary variables. That is, at i-th iterate, we sample $\alpha_k^{(i)}$ from the distribution $H(M, a_k^{(i)}, b_k^{(i)})$, where

$$a_k^{(i)} = M\frac{\psi(1+\alpha_k^{(i-1)})+\gamma}{2\alpha_k^{(i-1)}}$$

$$b_k^{(i)} = M(\psi(\alpha_0^{(i-1)})+\gamma) + \sum_{m=1}^{M}\psi(\alpha_k^{(i-1)}+n_{mk}) - \psi(\alpha_0^{(i-1)}+N_m)$$

[3] Need further study. This is equivalent to replacing the M-step in EM with a random sample from $e^{Q(\alpha)}$

		Setting A			Setting B		
		Squared error ($\times 0.001$)					
M	K	P-IG	P-IG(E)	MH	P-IG	P-IG(E)	MH
100	10	1.282	1.618	2.164	0.933	0.115	0.664
100	50	4.102	4.479	32.868	4.457	0.044	33.316
1000	10	0.437	0.222	0.769	0.862	0.028	0.333
1000	50	0.875	0.347	22.547	4.442	0.003	49.551
		Running time in seconds					
M	K	P-IG	P-IG(E)	MH	P-IG	P-IG(E)	MH
100	10	33.073	0.237	0.579	32.966	0.212	0.584
100	50	160.388	0.871	1.738	157.889	0.763	1.767
1000	10	346.532	1.593	5.675	365.227	1.59	5.663
1000	50	1368.643	5.944	17.652	1338.841	5.804	18.335
		Effective sample size ratio					
M	K	P-IG	P-IG(E)	MH	P-IG	P-IG(E)	MH
100	10	0.456	0.430	0.031	0.505	0.468	0.010
100	50	0.399	0.368	0.008	0.474	0.370	0.006
1000	10	0.389	0.369	0.014	0.495	0.458	0.009
1000	50	0.280	0.246	0.009	0.456	0.393	0.007

Discussion

- In phone sale example, we introduce $M \times K$ latent P-IG variables, one for each pair of $(\text{phone}_k, \text{region}_m)$. See the sampling algorithm for Multinomial-Dirichlet model (sample $\boldsymbol{\alpha}$ and $\boldsymbol{p}$ jointly) and the simulation study in our arXiv paper.

- The P-IG distribution class is flexible and admits fast and efficient stochastic simulation methods in widely used statistical models.

- It is possible to apply in regression setting:

$$\boldsymbol{p}_m | X, Z_m, \boldsymbol{\theta}_1, \boldsymbol{\theta}_2 \sim \text{Dirichlet}(X\boldsymbol{\theta}_1 + Z_m\boldsymbol{\theta}_2)$$

Thanks!